Computer Science & DA

Probability and Statistics

Permutation and Combination

Lecture No. 02

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Recap of previous lecture

Topic Permutation and Combination-01

Addition Rule: Either or / only one / Any one

Multi Rule → AND / Both / All

Combination → only selection is required $^nC_r = ?$

Permutation → Selection + Arrangement $^nP_r = {}^nC_r \times r!$

(*) Number of linear arrangements of n things at n places $= n!$

(*) RNA → ^nC_r, ^nP_r, $r!$; (RA) → only use M. Rule.

Topics to be Covered

Topic Permutation and Combination-02

(*) 5 flags & signals taking 3 at a time $\Rightarrow \underset{P_1}{5} \times \underset{P_2}{4} \times \underset{P_3}{3}$ ways $= {}^5C_3 \times 3! = {}^5P_3$

(RNA)

(*) Vegetables = 5 So Total different salads taking 3 Vegies at a time $= {}^5C_3$

(M-I) ~~Total~~ Salads $= {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5 = 31$

(M-II) Number of ways to deal with each Vegetable = 2 ways (either S or R)

Total " " " " " all " $= \underset{V_1}{2} \times \underset{V_2}{2} \times \underset{V_3}{2} \times \underset{V_4}{2} \times \underset{V_5}{2}$ ways $= 2^5$

$= 32$ ways

Various Salads: (SSSSS), (SSSSR), (SSSRR), (SSRRR), (SRRRR), (RRRRR)

${}^5C_5 = 1$, ${}^5C_4 = 5$, ${}^5C_3 = 10$, ${}^5C_2 = 10$, ${}^5C_1 = 5$, ${}^5C_0 = 1$ $= 32$

So Ans $= 31$.

Q. A person has 6 friends then in how many ways he/she can organize B'day party?

(M-I) Number of ways to deal with each f = 2 ways

" " " " " " all f = $2\times2\times2\times2\times2\times2$

$= 2^6 = 64$ ways.

But Req. Ans $= 2^6 - 1 = 63$ ways.

(M-II) He/she can organise B'day Party either by inviting = 1f or 2f or 3f or 4f or 5f or 6f

$= {}^6C_1 + {}^6C_2 + {}^6C_3 + {}^6C_4 + {}^6C_5 + {}^6C_6$

$= 6 + 15 + 20 + 15 + 6 + 1 = 63$

(iii) By inviting exactly 4f = ?

$= {}^6C_4 = 15$ ways

(iv) By inviting at least 3f = ?

$= {}^6C_3 + {}^6C_4 + {}^6C_5 + {}^6C_6$

(v) By inviting at most 2f = ?

$= {}^6C_1 + {}^6C_2 = 6 + 15 = 21$

(*) Number of linear arrangements of n different things = ? = $n!$

(*) Arrangement of alike items → Number of arrangements of n things in which p are alike, q are alike, r are alike and rest are different = ? = $\dfrac{n!}{p!\,q!\,r!}$

where $p+q+r+\underset{(?)}{\text{Rest}} = \underset{(n)}{\text{Total}}$

eg How many 4 digit Number can be formed using the digits $\overbrace{\underset{p}{\underline{6,6}},\ \underset{q}{\underline{8,8}}}^{\text{Toys}}$ (RNA)

(M-I) Various 4 digit NOS = (6688) (6868) (6886), (8866), (8686), (8668)

So Total 4 digit NOS = Six.

(M-II) $n = 4$

$p = 2,\ q = 2$

So Total arrangements $= \dfrac{4!}{2!\,2!} = 6$ Ans

Q: In how many ways 2 Maths Books ($= p$), 3 Physics Books ($= q$), 4 Chemistry ($= r$), 1 Hindi, 1 English, & 1 Geography Books can be arranged in a shelf?

Sol: $n = 12$, $p = 2$, $q = 3$, $r = 4$ So Total ways $= \dfrac{12!}{2!\,3!\,4!}$

Q: there are a copies of one book, b copies of two books, c copies of 3 books, and one-one copies of d books, then in how many ways we can arrange them in a shelf?

$\underbrace{HHH\ldots H}_{a}$, $\underbrace{EEE\ldots E}_{b}$, $\underbrace{PPP\ldots P}_{b}$, $\underbrace{GGG\ldots G}_{c}$, $\underbrace{CCC\ldots C}_{c}$, $\underbrace{SSS\ldots S}_{c}$,

$\underbrace{M, R, A, H, \ldots f}_{\text{Rest} = d}$

$n = (a + 2b + 3c + d)$ & $A_n = \dfrac{(a+2b+3c+d)!}{a!\,(b!)^2\,(c!)^3}$ //

Total ways of arranging Books $= \dfrac{n!}{\beta!\,\alpha!\,\gamma!\ldots\ldots} = \dfrac{(a+2b+3c+d)!}{a!\,b!\,b!\,c!\,c!\,c!}$

Q There are 6 periods (= Places) in each working day of a school. Then in how many ways we can arrange 5 different subjects (Persons) so that each subject gets at least one period?

(a) 1800
(b) 3600
(c) 720
(d) 120

eg

P	C	M	B	E	?=M
I	II	III	IV	V	VI

Total ways $= {}^5C_1 \times \dfrac{6!}{2!} = 1800$

Q. How many 6 digit Number can be formed using 1, 2, 3, 4 in which only one digit repeated thrice ?

(111 444)

Sol: Total 6 digit nos = (1 2 3 4 2 2) $\approx {}^4C_1 \times \frac{6!}{3!}$ = ✗

eg

(ii) in which there are exactly 2 pair of repeating digits ?

eg (1 2 3 4 3 1) = ${}^4C_2 \times \frac{6!}{2!\,2!}$

eg

$= 6$

M-II ${}^4C_1 \times {}^3C_1 \times \frac{6!}{2!\,2!}$ ✗

$= 12$

Concept of At least and at Most →

At least 3 App.

0 App or 1 App or 2 App or 3 App or 4 App or 5 Apples = Total possibilities.

At least one Apple.

At Most 2 App.

$\boxed{\text{At least one} = \text{Total} - \text{None}}$

Q: 3 dices are thrown simultaneously then find the possible number of outcomes in which at least one dice show digit '4' ?

Sol: Total Possible outcomes $= \underset{D_1}{6} \times \underset{D_2}{6} \times \underset{D_3}{6} = 216$

No. of ways in which No Dice show digit 4 $= \underline{5} \times \underline{5} \times \underline{5} = 125$

At least one dice show digit '4'

$= \text{Total} - \text{None} = 216 - 125$

$= 91$

Q. out of 6 Gents and 4 Ladies, a committee of 5 persons is to be formed. then find the Total number of committees if

① There is No Restriction = ? = ${}^{10}C_5$ = 252 Committees (this is Max Ans)

② At least 2L are there = ? = (2L & 3G) or (3L & 2G) or (4L & 1G) or (~~5L & 0G~~)

$= ({}^4C_2 \times {}^6C_3) + ({}^4C_3 \times {}^6C_2) + ({}^4C_4 \times {}^6C_1) = 186$?

③ At Most 2L are there = ? = (0L & 5G) or (1L & 4G) or (2L & 3G)

$= ({}^4C_0 \times {}^6C_5) + ({}^4C_1 \times {}^6C_4) + ({}^4C_2 \times {}^6C_3) = 186$

④ G are a Majority = ? = Same as Part ③ = 186

⑤ L " " " = ? = (4L & 1G) or (3L & 2G) $= ({}^4C_4 \times {}^6C_1) + ({}^4C_3 \times {}^6C_2) = 66$

(6) At least one L is there = ?

$(1L \,\&\, 4G)$ or $(2L \,\&\, 3G)$ or $(3L \,\&\, 2G)$ or $(4L \,\&\, 1G) = 246$

$= \text{Total} - \text{No Lady} = {}^{10}C_5 - \text{All G} = {}^{10}C_5 - {}^{6}C_5 = 246$

Analysis :- $(0L \,\&\, 5G)$ or $(1L \,\&\, 4G)$ or $(2L \,\&\, 3G)$ or $(3L \,\&\, 2G)$ or $(4L \,\&\, 1G) = \text{Total}$

$$({}^{4}C_0 \times {}^{6}C_5) + ({}^{4}C_1 \times {}^{6}C_4) + ({}^{4}C_2 \times {}^{6}C_3) + ({}^{4}C_3 \times {}^{6}C_2) + ({}^{4}C_4 \times {}^{6}C_1) = {}^{10}C_5$$

$$6 + 4 \times 15 + 6 \times 20 + 4 \times 15 + 1 \times 6 = 252$$

$$\underbrace{6 +}_{\text{At Most one L}} \underbrace{60 + 120 + 60 + 6}_{\text{At least one L} = 246} = 252$$

At least 2L = 186

6G & 4L

WRONG App of part (2) → At least 2L are there = ? = ${}^4C_2 \times {}^8C_3$ = ?

(M-III) Req Ans = $\underbrace{{}^4C_2 \times {}^6C_3 = 6 \times 30 = 180}$

$= 6 \times 56 = 336$

> Max Ans.

it is the answer different Question which is as follows:

" Find the No. of Committees in which there are exactly 2L ? "

Q. In a party there are 5 persons then find the Number of handshakes in that party?

Sol: M-I

$(P_1 P_2)$, $(\cancel{P_2 P_1})$ $(P_3 P_4)$
$(P_1 P_3)$ $(\cancel{P_2 P_2})$ $(P_3 P_5)$
$(P_1 P_4)$ $(P_2 P_3)$ $(P_4 P_5)$
$(P_1 P_5)$ $(P_2 P_4)$
$(P_2 P_5)$

Total Handshakes = 10

M-II Total Handshakes $= {}^5C_2 = 10$ Ans

Q. In a party there are 66 handshakes then find Number of persons in that party

Let Number of persons $= n$

ATQ, ${}^nC_2 = 66$

$$\frac{n(n-1)}{2} = 66$$

$n(n-1) = 132$

$= 12 \times 11$

$n(n-1) = 12(12-1)$

$n = 12$

De arrangements → when No one goes at right place assigned for him then such types of arrangements are called De-arrangements.

if there are n persons & n directed places then Total arrangements $= n!$

All Correct Arrangements $= 1$.

$$\text{All wrong Arrangements} = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots\cdots (-1)^n \frac{1}{n!}\right]$$

this Formula is Valid only when RNA.

eg there are 3 L & 3 addressed Envelopes then $\underbrace{\text{Total arrangements}}_{\text{RNA}} = 3! = 6$

$$\begin{pmatrix} E_1 & E_2 & E_3 \\ L_1 & L_2 & L_3 \end{pmatrix}, \begin{pmatrix} E_1 & E_2 & E_3 \\ L_1 & L_3 & L_2 \end{pmatrix}, \begin{pmatrix} E_1 & E_2 & E_3 \\ L_2 & L_1 & L_3 \end{pmatrix}, \begin{pmatrix} E_1 & E_2 & E_3 \\ L_2 & L_3 & L_1 \end{pmatrix} \begin{pmatrix} E_1 & E_2 & E_3 \\ L_3 & L_1 & L_2 \end{pmatrix} \begin{pmatrix} E_1 & E_2 & E_3 \\ L_3 & L_2 & L_1 \end{pmatrix}$$

All Correct.

All wrong.

(M-II) if $n=3$ then all wrong arrangements $= 3!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}\right] = 2$

Q: if 6 students are appearing in an exam where we have 6 directed chairs then find Total number of arrangements in which

(i) Everyone will be seated at Right Chair $= ? = 1$ way

(ii) No one will be seated on Correct Chair $= ? = 6!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}+\frac{1}{6!}\right]$

$= 265$

Note: Total arrangements $= 6! = 720$

(iii) At least two persons will be seated on wrong chair $= ? = 720-1 = 719$ Ans

Analysis:

for $n=2 \Rightarrow$ Dearrange $= 1$

for $n=3 \Rightarrow$ DeArrange $= 2$

" $n=4 \Rightarrow$ " " $= 9$

" $n=5 \Rightarrow$ " " $= 44$

" $n=6 \Rightarrow$ " " $= 265$

(All Correct) or (two wrong) or (three wrong) or (4 wrong) or (5 wrong) or (6 wrong) = Total Arrange

$$^6C_0 \times 1 + \underbrace{^6C_2 \times 1 + {}^6C_3 \times 2 + {}^6C_4 \times 9 + {}^6C_5 \times 44 + {}^6C_6 \times 265}_{\text{At least 2 wrong}} = 720$$

Q: There are 5 Balls and 5 distinct Cells then find the possible number of arrangements in which Ball B_i is not placed in Cell C_i $\forall i$ and No Cell Remains empty ?

??=(RNA)

(Ans = 44)

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THANK - YOU